A 200g ball rotates at a constant speed of 3 m/s on the end of 1.2 m long string. The

string describes a horizontal circle.

What is the centripetal acceleration of the object in m/s^2?

What is the centripetal force exerted on the object in N?

Select one:

a.

$3.33\ m/s^2$,0.67N

b.

$13.33\ m/s^2$,2.67N

c.

$7.5\ m/s^2$,1.5N

d.

$20.83\ m/s^2$,4.17N

الممسوحة ضوئيا بـ CamScanner

A simple pendulum of length L and mass M has frequency f. To increase its frequency to 5f:

Select one:

a.

Increase its length to 5L

b.

Increase its length to 25L

c.

Decrease its length to L/5

d.

Decrease its length to L/25

الممسوحة ضوئيا بـ CamScanner

A Ferrari is traveling in a uniform circular motion around a racetrack. What happens to the radial acceleration of the car if the velocity is doubled and the radius of the circle is doubled?

Select one:

a.

It increases by a factor of 4.

b.

It increases by a factor of 2.

c.

It remains the same.

d.

It increases by a factor of 8.

الممسوحة ضوئيا بـ CamScanner

Find the kinetic Energy of the ball (Mass = 3000g) when it is 8m above the ground in Joule:



Select one:

- a. **240**
- b. **210**
- c. **225**
- d. **450**

الممسوحة ضوئيا بـ CamScanner

The percentage error of the rotational mass:

The actual value of the rotational mass equals 14g, and m complete 10 revolutions

in 5 second, M = 75g,r = 30 cm, g = 10 m/s² .



Select one:

a.

11.4%

b.

12.86%

c.

9.2%

d.

15.3%

الممسوحة ضوئيا بـ CamScanner

A person exert a force on a string attached to a 325g ball to make the ball revolve

uniformly in a horizontal circle of radius 30cm. Find the tension in the string(in N) if

the ball makes two revolutions per second.

Select one:

a.

13.8

b.

15.38

c.

11.83

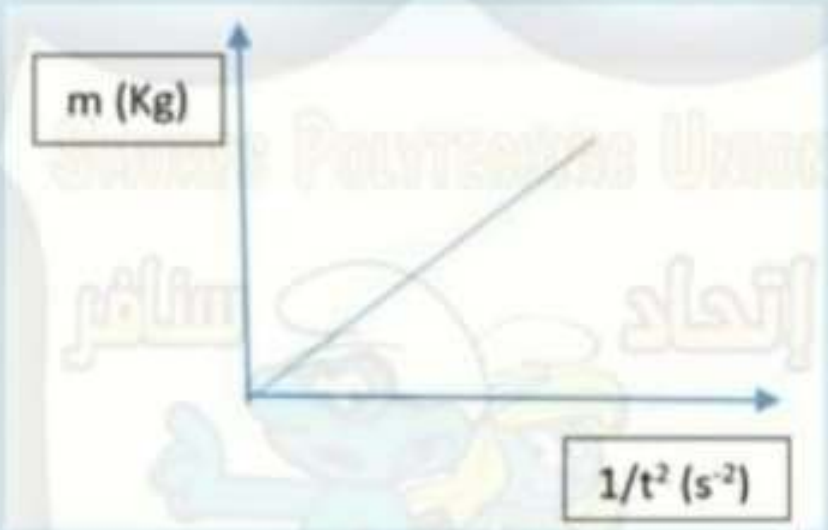
d.

18.93

الممسوحة ضوئيا بـ CamScanner

IN Newton's second law Experiment, the total mass equals 250g, the distance between

light gates equals 70 cm, Find acceleration of gravity in m/s^2 : (Slope = 0.041 $Kg.s^2$)



Select one:

- a. **8.5**
- b. **10.4**
- c. **9.8**
- d. **9.2**

الممسوحة ضوئيا بـ CamScanner

The period of vertical oscillation of a mass-spring system is T when the spring carries

a mass of 1.00 kg. What mass in (Kg) should be added to the 1.00 kg if the period is to

be increased to 1.50 T?

Select one:

a.

1.50

b.

1.25

c.

2.25

d.

1.00

الممسوحة ضوئيا بـ CamScanner

What does this metric micrometer read?



Select one:

- a.

  2.90 mm

- b.

  0.29 cm

- c.

  2.40 mm

- d.

  5.40 mm

الممسوحة ضوئيا بـ CamScanner

A simple pendulum is moved from the Earth ($g = 10\ m/s^2$) to the X planet. How does it

change the period of oscillations? Acceleration due to gravity on X planet = (5/3) m/s^2 :

Select one:

a.

The period is increased by factor √6

b.

The period is decreased by factor √6

c.

The period is decreased by factor four

d.

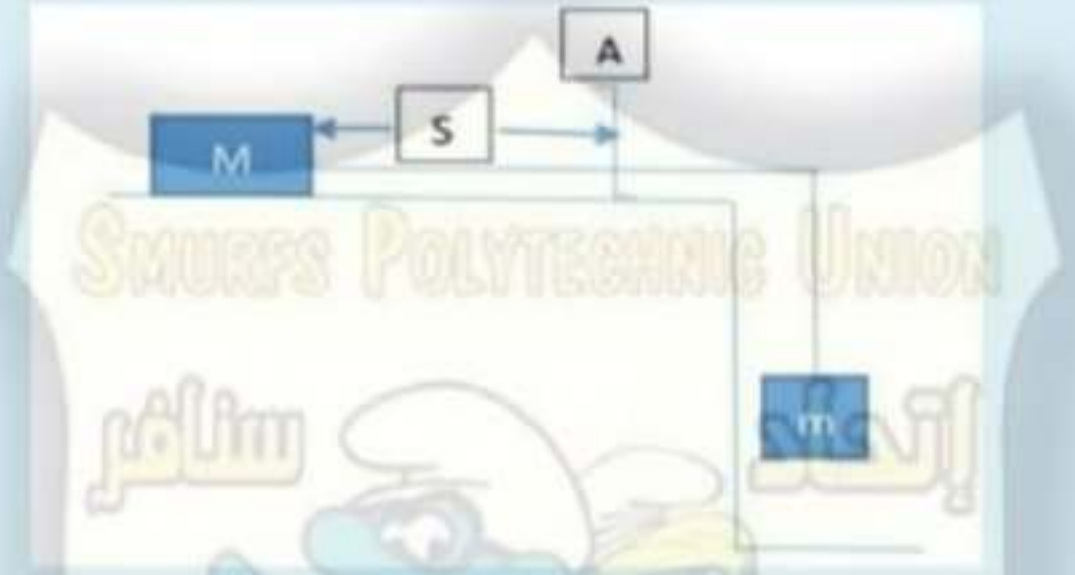
The period is increased by factor four

e.

The period remains the same

الممسوحة ضوئيا بـ CamScanner

In G.P.E. Experiment, M = 350g, m = 40g, L = 2 cm, t_A = 24.6 ms, Find S in cm:



Select one:

a.

21.8

b.

28.1

c.

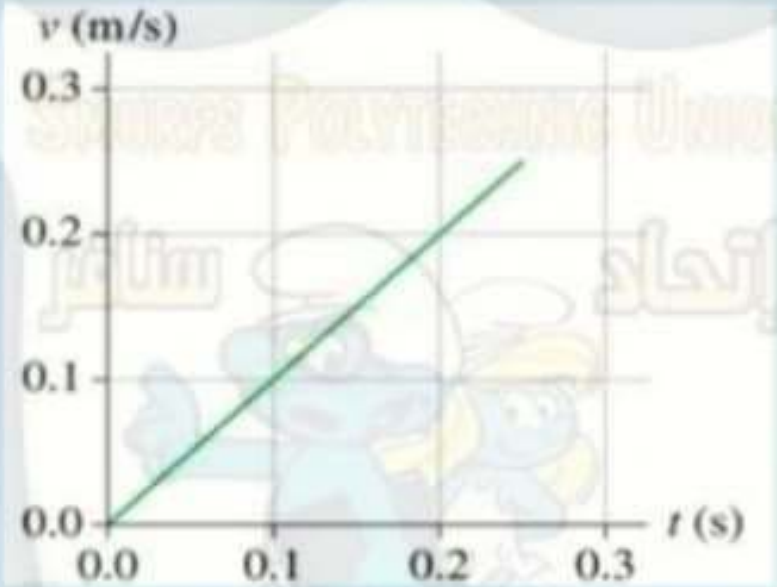
32.2

d.

64.4

الممسوحة ضوئيا بـ CamScanner

Scallops eject water from their shells to provide a thrust force. The graph shows a smoothed graph of actual data for the initial motion of a 25 g scallop speeding up to escape a predator. What is the magnitude of the net force(N) needed to achieve this motion?



Select one:

- a.

  0.025

- b.

  05

- c.

  2.5

- d.

  0.005

الممسوحة ضوئيا بـ CamScanner

A 23 kg cart (A) moving at 8 m/s collides with a 16 kg cart(B) moving 13 m/s. If the

carts stick together after the collision, what is the velocity after the collision?



Select one:

a.

- 1.04 m/s

b.

- 0.62 m/s

c.

0.6 m/s

d.

10.05 m/s

الممسوحة ضوئيا بـ CamScanner

A 1250 kg car is moving down the highway with a velocity of 38.0 m/s when it bumps into the car ahead of

its which has a mass of 900 kg and a velocity of 30.0 m/s. After the collision, the two cars stick together.

What will be the resulting velocity of the two cars together? How much energy will be lost in this collision?

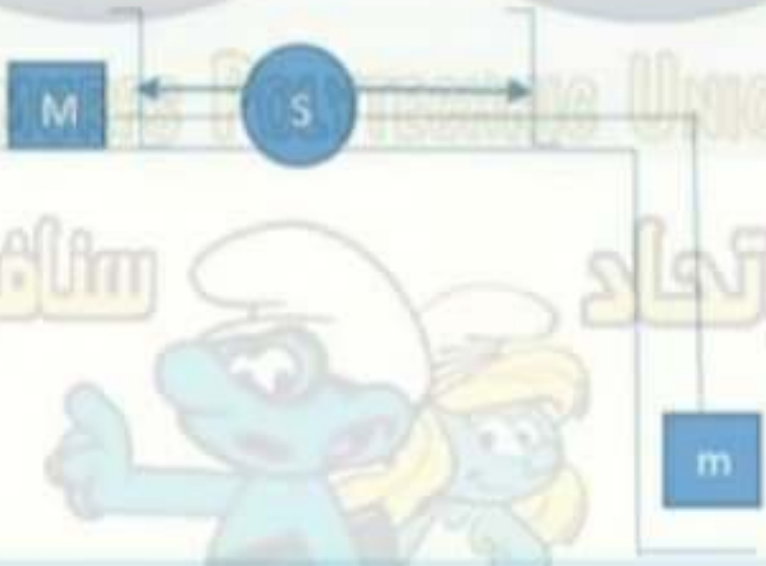
Select one:

- a. (32.7 m/s, 16879.4 J)
- b. (29.1 m/s, 13702 J)
- c. (34.65 m/s, 16830.8 J)
- d. (30.2 m/s, 23307 J)

الممسوحة ضوئيا بـ CamScanner

The time required to cut the distance between the two photogates in second:

(M = 220 g , m = 30 g m , S = 70 cm , g = 10 m/s^2)



Select one:

a.

1.97

b.

1.08

c.

1.33

d.

0.95

الممسوحة ضوئيا بـ CamScanner

A 2500 N Car is supported by 5 springs. If each spring is deflected by 4 cm what is the

spring constant in (N/m)?

Select one:

- a. **62500**
- b. **500**
- c. **12500**
- d. **125**

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الممسوحة ضوئيا بـ CamScanner

Three simple pendulums, with masses m_1, $m_2 = 2m_1$, and $m_3 = 3m_1$ hung at the end,

have the same shape and size and are suspended at the same point. Rank them

according to their periods, from shortest to longest.

Select one:

- a. All three are the same
- b. 2, 1, 3
- c. 3, 2, 1
- d. 1, 2, 3

CamScanner الممسوحة ضوئيا بـ

An object of mass m_1 = 4 Kg, moving with velocity V_{i1} = 12 m/s, collides head-on with a stationary object

whose mass is m_2 = 8 Kg. Given that the collision is elastic, what are the final velocities of the two objects

(V_{1f},V_{2f})? Neglect friction.

Select one:

a.

(-6, 6) m/s

b.

(-4, 8) m/s

c.

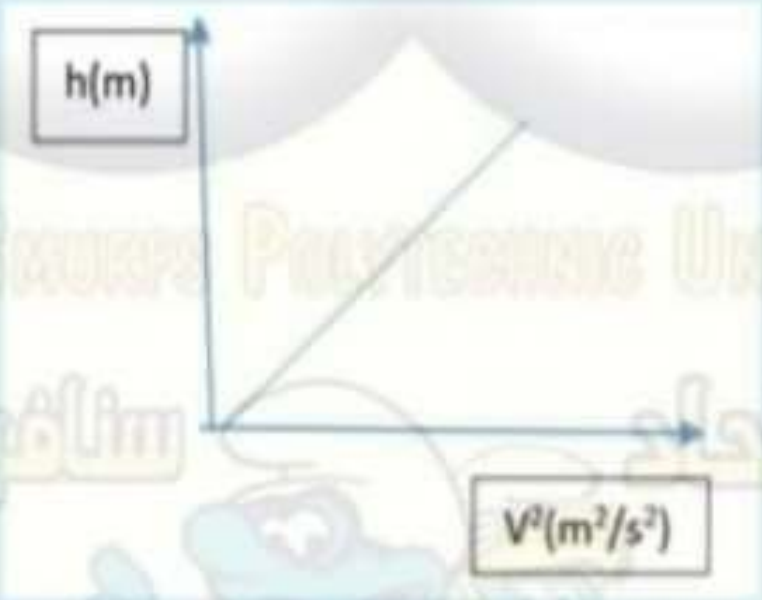
(-12, 12) m/s

d.

(7.2, 2.4) m/s

الممسوحة ضوئيا بـ CamScanner

In G.P.E. Experiment, M = 200g, m = 20g, Slope = 0.65 s^2/m, Find the acceleration of gravity in (m/s^2):



Select one:

a.

12

b.

9.8

c.

11

d.

8.87

الممسوحة ضوئيا بـ CamScanner

Find:

$$4.7832 + 1.234 + 2.02 =$$

(Using significant figure rules)

Select one:

- a. **8.04**
- b. **8.0372**
- c. **8.037**
- d. **8.0**

الممسوحة ضوئيا بـ CamScanner

A cylinder hangs motionless 20 cm below its equilibrium position from a spring with a

constant of 50 N/m. What is the mass of the cylinder?



Select one:

a.

1

b.

10

c.

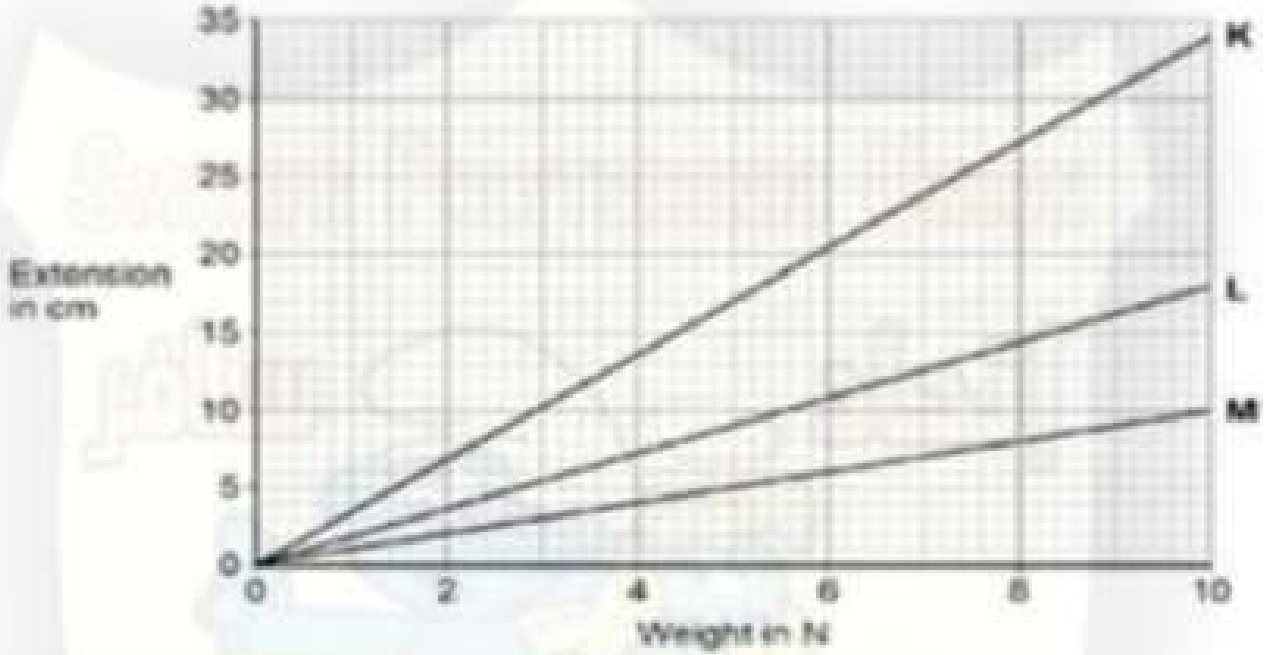
100

d.

25

CamScanner الممسوحة ضوئيا بـ

Which out of springs K, L or M from the graph have the highest spring constant.



Select one:

a.

M

b.

Same spring constant

c.

K

d.

L

الممسوحة ضوئيا بـ CamScanner

Find the total mass of the system:

(S= 75 cm, g = 10 m/s^2, Slope = 3.03 s^2/N)



Select one:

- a. 150
- b. 250
- c. 220
- d. 180

الممسوحة ضوئيا بـ CamScanner